STPM 2015 PHYSICS 960/1

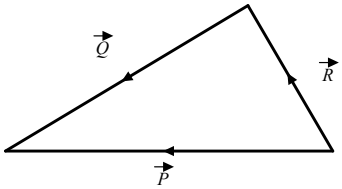
Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.

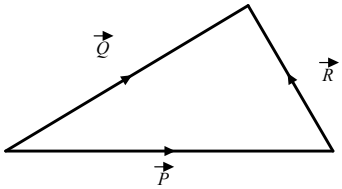
Section A [15 marks]

1. Which vector diagram shows $\boldsymbol{P} = \boldsymbol{Q} - \boldsymbol{R}$?

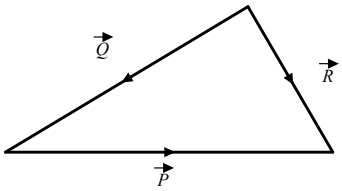
A



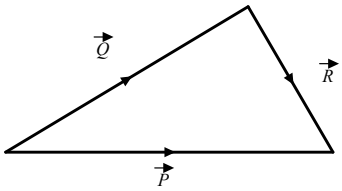
B



C



D



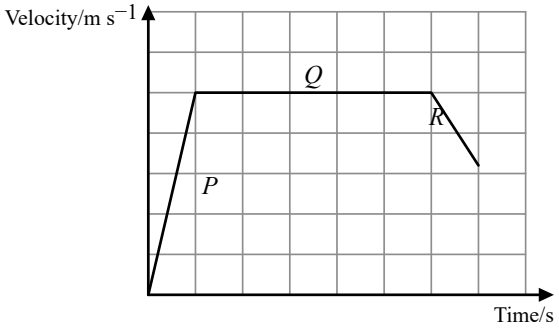
Answer: B

$$\boldsymbol{P} = \boldsymbol{Q} - \boldsymbol{R} \Leftrightarrow \boldsymbol{P} + \boldsymbol{R} = \boldsymbol{Q}$$

So $\boldsymbol{P}$ and $\boldsymbol{R}$ are drawn head to tail, and $\boldsymbol{Q}$ joins the tail of $\boldsymbol{P}$ to the head of $\boldsymbol{R}$, as in B.

Explanation: Rearrange the subtraction into an addition. In a vector triangle, the vector that goes from the start point to the end point is the sum of the other two.

2. The variation of velocity with time of a car travelling along a straight road is shown in the graph below.



Which statement about the force acting on the car is **not** true?

A A nett force acts on the car during interval *P*.
B No nett force acts on the car during interval *Q*.
C Opposing forces may be acting on the car during interval *Q*.
D The magnitude of the nett force acting during interval *P* is less than that during interval *R*.

Answer: D

The force is proportional to the acceleration, the gradient of the v–t graph.

The gradient in P is steeper than in R, so the net force in P is **greater**. Statement D is not true.

Explanation: In Q the velocity is constant, so the net force is zero, although driving force and friction may both act and balance (B and C are true).

3. The nett force on an object moving in a straight line on a frictionless surface suddenly becomes zero. The object will then

A stop instantly
B require a short time to stop
C slow down gradually
D move at constant velocity

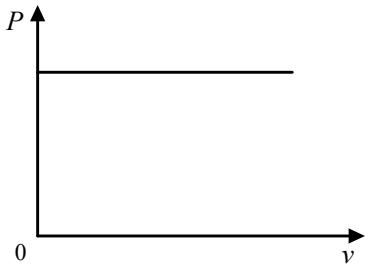
Answer: D

With zero net force the object has no acceleration (Newton's first law), so it continues to move at constant velocity.

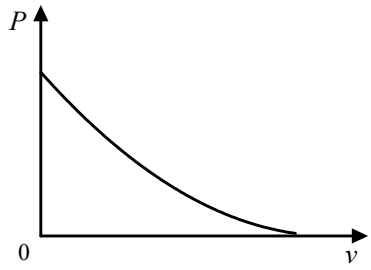
Explanation: The surface is frictionless, so there is nothing to slow the object down.

4. A car moves with uniform acceleration from rest. Which graph shows the variation of power P of the car with velocity v?

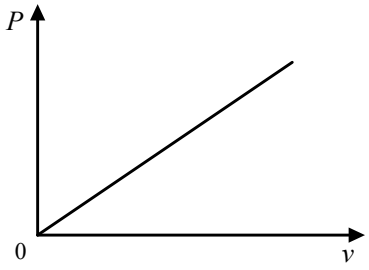
A



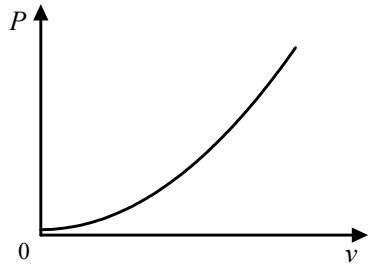
B



C



D



Answer: C

$$F = ma \text{ is constant; } P = Fv \Longrightarrow P \propto v$$

A straight line through the origin.

Explanation: Uniform acceleration needs a constant driving force (ignoring resistance), so the power increases in proportion to the speed.

5. A particle of mass 0.020 kg is attached to an elastic string which is fixed at one end on a smooth horizontal table. The length of the string is 0.50 m with force constant 40 N m^{-1}. When the particle is moving in a horizontal circle of radius 0.65 m, what is the angular velocity of the particle?

A 21 rad s^{-1} B 24 rad s^{-1} C 45 rad s^{-1} D 51 rad s^{-1}

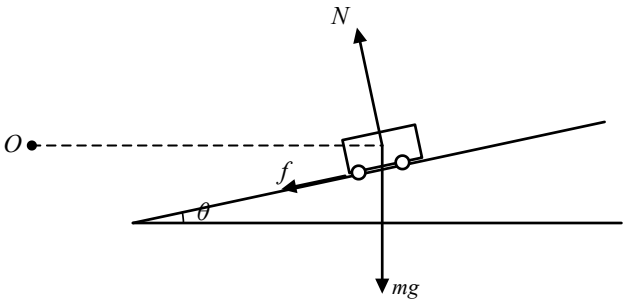
Answer: A

Extension = 0.65 − 0.50 = 0.15 m; tension $T = (40)(0.15) = 6.0$ N

$$T = mr\omega^2 \Longrightarrow \omega = \sqrt{\left(\frac{6.0}{(0.020)(0.65)}\right)} = 21 \text{ rad s}^{-1}$$

Explanation: The tension in the stretched string provides the centripetal force. Use the radius of the circle (0.65 m), not the natural length.

6. A car of mass m is moving along a sideways slope with centre O as shown in the diagram below.



If N is the normal reaction on the car and f is the friction on the tyres, the centripetal force on the car is

A $f + mg \sin\theta$ B $f + mg \cos\theta$
C $N \sin\theta + f \cos\theta$ D $N \cos\theta + f \sin\theta$

Answer: C

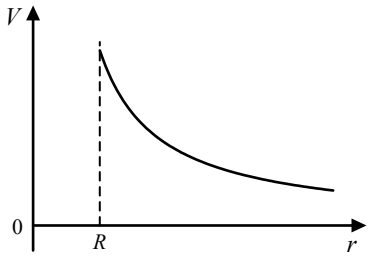
The centre O is horizontally level with the car, so the centripetal force is the sum of the horizontal components of the forces towards O:

$$F = N \sin\theta + f \cos\theta$$

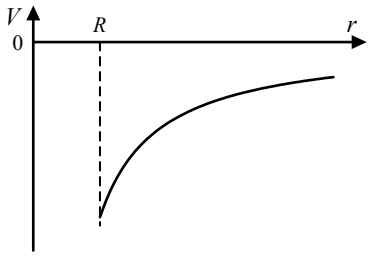
Explanation: The weight is vertical, so it has no horizontal component. N is perpendicular to the slope (horizontal component $N \sin\theta$) and f acts along the slope (horizontal component $f \cos\theta$).

7. Which graph represents the variation of gravitational potential V with distance r from the centre of the Earth for region $r > R$?
[R is the radius of the Earth.]

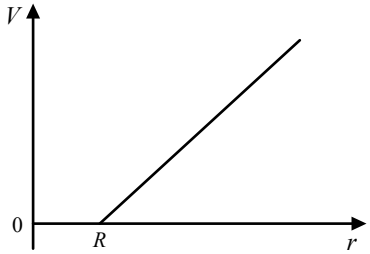
A



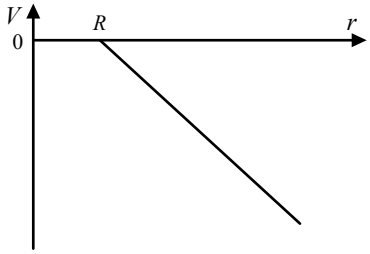
B



C



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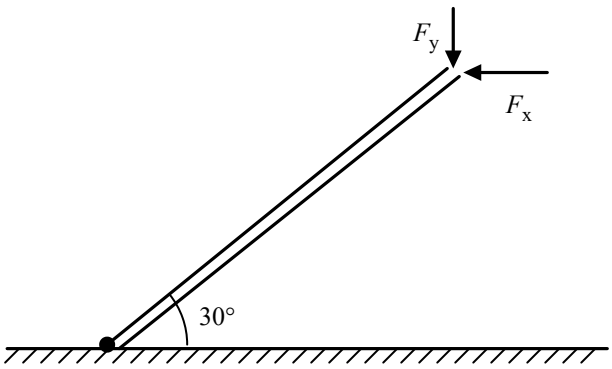
Answer: B

$$V = -\frac{GM}{r}$$

Negative and increasing towards zero as r increases (a $-1/r$ curve), starting at $r = R$.

Explanation: Gravitational potential is zero at infinity and negative everywhere else.

8. A light rod is hinged at an angle of 30° to the horizontal and is kept in equilibrium by the forces F_y and F_x as shown in the diagram below.



If F_y is 200 N, what is F_x?

A 100 N B 115 N C 173 N D 346 N

Answer: D

Moments about the hinge: $F_y(L \cos 30°) = F_x(L \sin 30°)$

$$F_x = \frac{200}{\tan 30°} = 346 \text{ N}$$

Explanation: The perpendicular distance from the hinge to the line of the vertical force is L cos 30°, and to the horizontal force is L sin 30°.

9. Which property of material is **not** due to plastic deformation?

A Brittle
B Ductile
C Ability to bend
D Ability to withstand stress

Answer: A

A brittle material breaks without (or with very little) plastic deformation.

Explanation: Ductility, the ability to bend and the ability to withstand stress beyond the elastic limit all depend on plastic deformation.

10. The density of an ideal gas of molar mass M at pressure p and temperature T is

A $\frac{p}{MRT}$ B $\frac{MRT}{p}$ C $\frac{pM}{RT}$ D $\frac{RT}{pM}$

Answer: C

$$pV = \frac{m}{M}RT \Longrightarrow \rho = \frac{m}{V} = \frac{pM}{RT}$$

Explanation: Replace the number of moles by mass/molar mass, then rearrange for mass/volume.

11. What is the r.m.s. speed of oxygen molecules in the air at 20 °C?
[Molar mass of oxygen is 32.0 g.]

A 478 m s^{-1} B 617 m s^{-1} C 676 m s^{-1} D 872 m s^{-1}

Answer: A

$$c_{\text{r.m.s.}} = \sqrt{\left(\frac{3RT}{M}\right)} = \sqrt{\left(\frac{3(8.31)(293)}{0.0320}\right)} = 478 \text{ m s}^{-1}$$

Explanation: Use the molar mass in kg mol^{-1} (0.0320) and the temperature in kelvin.

12. Five moles of an ideal gas is compressed at a constant pressure and its temperature increases from 300 K to 450 K. The work done in this process is

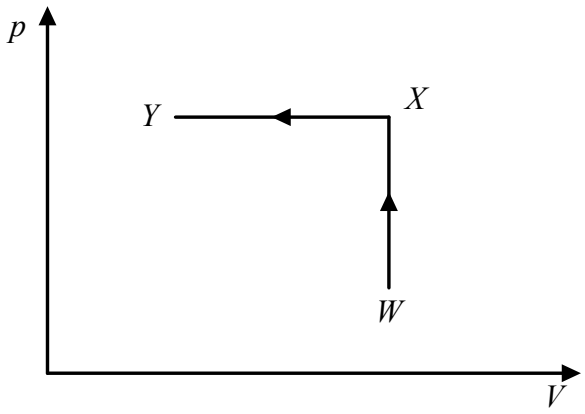
A 1.2×10^3 J B 6.2×10^3 J C 1.2×10^4 J D 1.6×10^4 J

Answer: B

$$|W| = p|\Delta V| = nR|\Delta T| = (5)(8.31)(150) = 6.2 \times 10^3 \text{ J}$$

Explanation: At constant pressure, $p\Delta V = nR\Delta T$. (A compression at constant pressure would actually lower the temperature; the magnitude of the work is the same either way.)

13. A process for an ideal gas from state $W \rightarrow X \rightarrow Y$ is shown in the p-V diagram below.



If the internal energy of the gas at W and Y are the same, which is true about the net heat transfer and the net work done for the whole process?

	Net heat transfer	*Net work done*
A	Released by the gas	On the gas
B	Released by the gas	By the gas
C	Absorbed by the gas	On the gas
D	Absorbed by the gas	By the gas

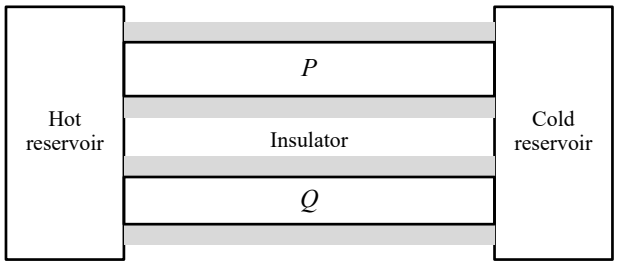
Answer: A

$W \rightarrow X$: constant volume, no work. $X \rightarrow Y$: compression, work done **on** the gas.

$U_W = U_Y \Longrightarrow \Delta U = 0 \Longrightarrow Q = W_{\text{by gas}} < 0$: heat is released by the gas.

Explanation: Apply the first law to the whole process. With no change in internal energy, the energy received as work must leave as heat.

14. Two well-insulated metal rods P and Q of the same length with their ends placed in hot and cold reservoirs are shown in the diagram below.



The thermal conductivity of P is twice that of Q. If the cross-sectional area of P is one-quarter that of Q, which statement is true of the rods after a steady state is achieved?

A The rates of heat flow in P and Q are the same.
B The rate of heat flow in P is twice that in Q.
C The temperature gradients in P and Q are the same.
D The temperature gradient in P is twice that in Q.

Answer: C

$$\text{Temperature gradient} = \frac{\Delta\theta}{L}\text{: same ends and same length} \Longrightarrow \text{same gradient}$$

$$\text{Rates: } \frac{H_P}{H_Q} = \frac{2k(A/4)}{kA} = \frac{1}{2}$$

Explanation: Both rods are insulated and connect the same two reservoirs, so the temperature drop per unit length is the same. The rate of heat flow in *P* is half that in *Q*.

15. An insulated aluminium rod and copper rod have the same cross-sectional area and temperature difference between their ends. The thermal conductivities of aluminium and copper are 240 W m^{-1} K^{-1} and 400 W m^{-1} K^{-1} respectively. If the rates of heat flow in the aluminium and copper rods are the same, what is the ratio of the length of the aluminium rod to the length of the copper rod?

A 3 : 8 B 3 : 5 C 5 : 3 D 8 : 3

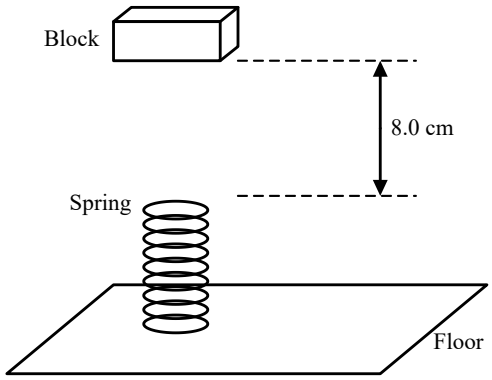
Answer: B

$$\frac{k_{Al}A\Delta\theta}{L_{Al}} = \frac{k_{Cu}A\Delta\theta}{L_{Cu}} \Longrightarrow \frac{L_{Al}}{L_{Cu}} = \frac{240}{400} = \frac{3}{5}$$

Explanation: For equal rates of heat flow, the poorer conductor (aluminium) must be shorter.

Section B [15 marks]

16. A block of mass 65 g is placed at 8.0 cm above the top end of a long spring which is fixed on the floor as shown in the diagram below.



The block is released and falls vertically onto the spring.
[The spring constant is 66 N m^{-1}.]

(a) State the energy transformation of the system from the point of release to maximum compression of the spring. [3]

Answer

As the block falls: gravitational potential energy → kinetic energy	1
On the spring: kinetic energy and gravitational potential energy → elastic potential energy of the spring	1
At maximum compression: all the energy is stored as elastic potential energy (kinetic energy = 0)	1
	3

Explanation: The block keeps losing gravitational potential energy while it compresses the spring, so this energy is also stored in the spring.

(b) Determine the maximum compression of the spring. [4]

Answer

$mg(h + x) = \frac{1}{2}kx^2$	1
$(0.065)(9.81)(0.080 + x) = \frac{1}{2}(66)x^2$	1
$33x^2 - 0.638x - 0.0510 = 0$	1
$x = 0.050$ m	1
	4
Total for question	**7**

Explanation: The total fall is 8.0 cm plus the compression x. Take the positive root of the quadratic.

17. (a) Describe *degree of freedom* in a gas. [2]

Answer

The degree of freedom is the number of independent ways in which a molecule can possess energy (independent modes of motion)	1
e.g. translational, rotational and vibrational motion; a monatomic molecule has 3 (translational)	1
	2

Explanation: A diatomic molecule at room temperature has 5: 3 translational and 2 rotational.

(b) Using the principle of equipartition of energy, derive an expression for the kinetic energy K, in terms of temperature T and degree of freedom f, of a gas containing N molecules. [3]

Answer

Equipartition: each degree of freedom has an average energy of $\frac{1}{2}kT$ per molecule — 1

One molecule with f degrees of freedom: $\frac{f}{2}kT$ — 1

N molecules: $K = \frac{f}{2}NkT$ — 1

3

Explanation: Start the derivation from the principle of equipartition of energy and build up from one degree of freedom to the whole gas.

(c) Nitrogen gas of 3.0 moles is kept in a fixed container at a temperature of 100 K. It is then heated to a temperature of 600 K. Determine the change in internal energy of the gas. [3]

Answer

At 100 K only translational motion is taken ($f = 3$): $U_1 = \frac{3}{2}(3.0)(8.31)(100) = 3.74 \times 10^3$ J — 1

At 600 K rotation is included ($f = 5$): $U_2 = \frac{5}{2}(3.0)(8.31)(600) = 3.74 \times 10^4$ J — 1

$\Delta U = 3.74 \times 10^4 - 3.74 \times 10^3 = 3.37 \times 10^4$ J — 1

[If $f = 5$ is used at both temperatures, $\Delta U = \frac{5}{2}(3.0)(8.31)(500) = 3.12 \times 10^4$ J]

3

Total for question **8**

Explanation: The intended point is that the number of degrees of freedom of a diatomic gas increases with temperature, so the internal energy must be found separately at each temperature, not from one value of f.

Section C [30 marks]

18. (a)

(i) State Newton's second and third laws of motion. [2]

Answer

Second law: the rate of change of momentum of a body is directly proportional to the resultant force and takes place in the direction of the force — 1

Third law: if body A exerts a force on body B, body B exerts a force of equal magnitude in the opposite direction on body A (action and reaction) — 1

2

Explanation: The action–reaction pair acts on different bodies, so they do not cancel.

(ii) A 4500 kg rocket is fixed for vertical launching. The burnt fuel is ejected from the rocket at a speed of 850 m s^{-1}. Calculate the mass of fuel ejected per second so that the rocket starts to move with an acceleration of 7.5 m s^{-2}. [4]

Answer

Resultant force: $F - mg = ma$ 1

$F = (4500)(9.81 + 7.5) = 7.79 \times 10^4$ N 1

Thrust $F = v\frac{dm}{dt}$ 1

$\frac{dm}{dt} = \frac{7.79 \times 10^4}{850} = 91.6$ kg s^{-1} 1

4

Explanation: The thrust (rate of change of momentum of the ejected fuel) must overcome the rocket's weight as well as provide its acceleration.

(b)

(i) Define *coefficient of static friction*. [1]

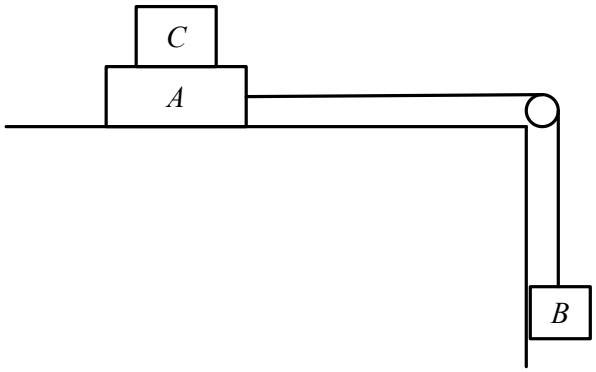
Answer

The ratio of the limiting (maximum) static friction to the normal reaction 1

1

Explanation: $\mu_s = f_{max}/R$; it has no unit.

(ii) Block *A* on a horizontal surface of a table is connected to block *B* by an inelastic string passing over a light and frictionless pulley. Block *C* is put on top of block *A* as shown in the diagram below.



The masses of block A and B are 4.0 kg and 2.2 kg respectively. The coefficient of static friction between block A and the table is 0.25. Determine the minimum mass of block C to prevent block A from sliding. [3]

Answer

$T = m_B g = (2.2)(9.81) = 21.6$ N — 1

Limiting friction on A: $\mu_s(m_A + m_C)g = 21.6 \Longrightarrow 0.25(4.0 + m_C)(9.81) = 21.6$ — 1

$m_C = 4.8$ kg — 1

3

Explanation: Block C increases the normal reaction on A, and so the maximum friction. At the minimum mass, friction is at its limiting value.

(iii) If block C is lifted, draw a diagram of forces acting on block A and calculate the acceleration of the blocks.
[The coefficient of kinetic friction between block A and the table is 0.20.] [5]

Answer

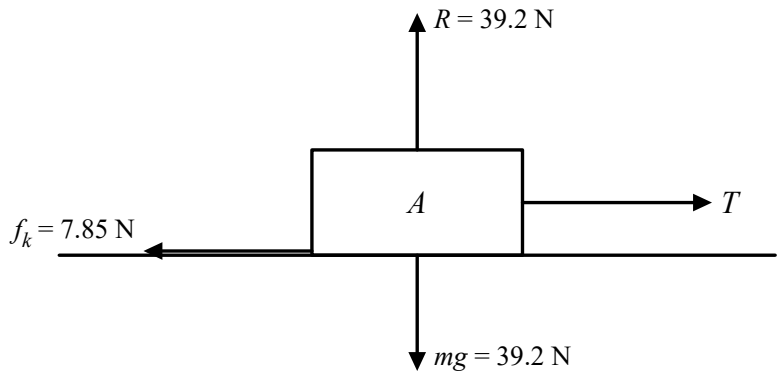


Diagram: R up, mg down, T towards the pulley, kinetic friction f_k opposite — 2

Block A: $T - 0.20(4.0)(9.81) = 4.0a$; block B: $(2.2)(9.81) - T = 2.2a$ — 1

Adding: $21.58 - 7.85 = 6.2a$ — 1

$a = 2.22$ m s^{-2} — 1

5

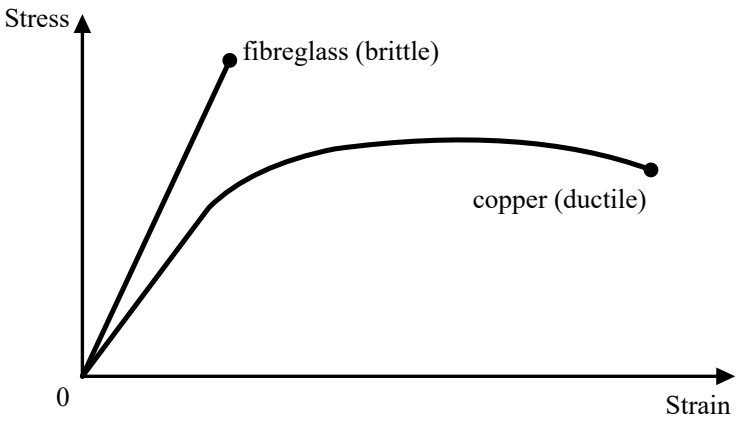
Total for question **15**

Explanation: Once A slides, kinetic friction (coefficient 0.20) acts. Treat the blocks separately and add the equations to eliminate the tension.

19. (a) A copper wire and fibreglass of the same dimensions are each suspended from a retort stand. Each of them is then slowly stretched by adding loads to their free ends until they break.

(i) On the same axes, sketch graphs of stress against strain to show the two materials. [2]

Answer



Fibreglass: steep straight line ending suddenly (no plastic region)	1
Copper: smaller gradient, with a long plastic region before breaking	1
	2

Explanation: Both graphs must be on the same axes so that the gradients and breaking points can be compared.

(ii) From your graphs in (a)(i), state **three** different characteristics of the materials. [3]

Answer

• Fibreglass has a greater Young's modulus (stiffer) – steeper gradient	1
• Fibreglass is brittle while copper is ductile	1
• Fibreglass has a greater breaking stress / copper breaks at a much larger strain	1
	3

Explanation: Each characteristic should be read directly from the two graphs; do not contradict your own sketch.

(b) A copper wire with initial length of 0.755 m and cross-sectional area of 3.75×10^{-7} m^2 is stretched until the strain in the wire is 0.015.
[Young's modulus of copper is 1.30×10^{11} Pa.]

(i) Calculate the force applied. [3]

Answer

Stress = $E\varepsilon = (1.30 \times 10^{11})(0.015) = 1.95 \times 10^{9}$ Pa — 1

F = stress × $A = (1.95 \times 10^{9})(3.75 \times 10^{-7})$ — 1

$F = 731$ N — 1

3

Explanation: Young's modulus links stress and strain; the force is stress times the cross-sectional area.

(ii) Calculate the strain energy in the wire. [3]

Answer

$e = \varepsilon L = (0.015)(0.755) = 1.13 \times 10^{-2}$ m — 1

$U = \frac{1}{2}Fe = \frac{1}{2}(731)(1.13 \times 10^{-2})$ — 1

$U = 4.14$ J — 1

3

Explanation: The strain energy is the area under the force–extension graph (assuming Hooke's law holds).

(iii) If the cross-sectional area of the wire changes but the volume remains the same, determine the fractional change in cross-sectional area of the wire. [4]

Answer

Volume constant: AL = constant — 1

$\frac{\Delta A}{A} + \frac{\Delta L}{L} = 0$ — 1

$\frac{\Delta A}{A} = -\frac{\Delta L}{L} = -0.015$ — 1

The cross-sectional area decreases by a fraction 0.015 (1.5%) — 1

4

Total for question **15**

Explanation: For small changes, the fractional changes in area and length add up to the fractional change in volume, which is zero.

20. (a)

(i) Define *molar heat capacity* at constant volume of an ideal gas. [1]

Answer

The quantity of heat required to raise the temperature of one mole of the gas by 1 K at constant volume — 1

1

Explanation: Unit: J mol^{-1} K^{-1}.

(ii) Using the first law of thermodynamics and the ideal gas law, show that $C_{p,m} = C_{V,m} + R$, where $C_{p,m}$ is the molar heat capacity at constant pressure, $C_{V,m}$ is the molar heat capacity at constant volume and R is the gas constant. [6]

Answer

First law: $Q = \Delta U + W$ — 1

Constant volume: $W = 0$, so $nC_{V,m}\Delta T = \Delta U$ — 1

ΔU depends only on ΔT for an ideal gas, so it is the same at constant pressure — 1

Constant pressure: $nC_{p,m}\Delta T = \Delta U + p\Delta V$ — 1

Ideal gas law: $p\Delta V = nR\Delta T$ — 1

$nC_{p,m}\Delta T = nC_{V,m}\Delta T + nR\Delta T \Longrightarrow C_{p,m} = C_{V,m} + R$ — 1

6

Explanation: The extra heat needed at constant pressure is the work done by the gas as it expands, $nR\Delta T$.

(b) Describe the change in the thermodynamic quantities of an adiabatic compression according to the first law of thermodynamics. [3]

Answer

No heat is exchanged: $Q = 0$ — 1

Work is done on the gas (the work done by the gas is negative), so $\Delta U = -W$ is positive — 1

The internal energy, and so the temperature, of the gas increases — 1

3

Explanation: In an adiabatic compression all the work done on the gas goes into its internal energy.

(c) A gas with $\gamma = 1.40$, initially at temperature 30.0 °C, volume 3.50×10^{-3} m^3 and pressure 1.013×10^5 Pa, undergoes an adiabatic compression to one third of its original volume. Assume the gas behaves as an ideal gas.

(i) State the type of gas molecules used. [1]

Answer

Diatomic gas molecules ($\gamma = 1.40 = \frac{7}{5}$) 1

1

Explanation: $\gamma = (f + 2)/f = 1.40$ gives $f = 5$.

(ii) Calculate the final pressure of the gas. [2]

Answer

$p_2 = p_1(\frac{V_1}{V_2})^{\gamma} = (1.013 \times 10^5)(3)^{1.40}$ 1

$p_2 = 4.72 \times 10^5$ Pa 1

2

Explanation: The volume is reduced to one third, so the ratio $V_1/V_2 = 3$.

(iii) Determine the amount of work done on the gas. [2]

Answer

$W_{on} = \frac{p_2V_2 - p_1V_1}{\gamma - 1} = \frac{(4.716 \times 10^5)(1.167 \times 10^{-3}) - (1.013 \times 10^5)(3.50 \times 10^{-3})}{0.40}$ 1

$W_{on} = 489$ J 1

2

Total for question **15**

Explanation: The work done on the gas equals its increase in internal energy. (The work done by the gas is −489 J.)